

# Week 3: Parameter Estimation, Hypothesis Testing, and Decisions

DSAA1001 · Three-tier learning sheet · Advanced Learning Sheet

**Opening diagnostic challenge.** You observed 20 past outdoor event days; 8 had rain. Your teammate says: “The rain probability is 40%, so if our threshold is 30%, move the event indoors.”

What is right about this rule, and what is missing before it becomes a defensible statistical decision?

## Reading contract and capabilities

### Week orientation

**Backbone question.** When data are uncertain, how do we move **from observations to an estimate**, from an estimate to a test, and from a test or probability to an action – without confusing **“what the model says”** with “what the world guarantees”?

**Running example.** A campus operations team must decide whether to move an outdoor orientation event indoors. The main data record whether it rained on each past event day. A separate numerical-score example is used for Gaussian estimation. The first unknown is  $p$ , the probability of rain on an event day; operational costs enter later as inputs to a decision, not as observations.

A **statistical procedure** will mean a **rule that takes data and returns** an estimate, a test decision, a prediction, or an action.

### Learning objectives

- formulate a parameter-estimation problem from data and a model;
- compute and interpret maximum-likelihood estimates for Bernoulli and Gaussian models;
- explain how Bayesian estimation combines a prior with likelihood;
- compare MLE, MAP, and posterior-mean estimators through their decision roles;
- evaluate estimators using bias, variance, MSE, and consistency;
- formulate basic hypothesis tests, p-values, Type I/II errors, and likelihood-ratio tests;
- choose actions under uncertainty using loss, expected loss, and predictive distributions;
- critique when a mathematically correct estimate or test still fails the operational question.

### Breadcrumb Prompt

#### Write your starting rule of thumb

In one sentence, write how you currently turn past observations such as “8 rainy days out of 20” into an operational decision. At the end of the sheet, revise the rule so it separates estimation, evidence against a claim, prediction, and action cost.

- **Tier 1: fast path.** Estimate, test, decide: what each object is for.
- **Tier 2: formal machinery.** Likelihoods, posteriors, estimator properties, p-values, likelihood ratios, and expected loss.
- **Tier 3: construction and comparison.** Build a suitable target, compare procedures, and connect evidence to decisions.

### Pause and Think

**Question:** If the estimated rain probability is 40%, should the team automatically move indoors?

*Hint:* Ask what the team loses under each wrong decision. A 40% chance is high for a catastrophic loss and low for a mild inconvenience.

**Answer:**

No. Probability is only one input. The action also depends on loss: cost of moving indoors, cost of being rained out, reputational damage, and whether more information can be purchased before deciding.

**Remark. Notation map.** Data are  $D$  or  $X_1, \dots, X_n$ . Parameters are  $\theta$  in general and  $p, \mu, \sigma^2$  in examples. An **estimator** is a random variable (a rule applied to data) such as  $\hat{\theta}(X_1, \dots, X_n)$ ; an **estimate** is the numeric value after data are observed. A **likelihood** is  $L(\theta) = P(D \mid \theta)$  or a density value for continuous data; it measures how plausible the observed data would be if parameter  $\theta$  were true. A hypothesis test returns reject / do not reject; a decision rule returns an action.

# 1. The pipeline: model, estimate, test, decide

The campus team does not really want a formula for its own sake. It wants a **defensible action** before the event. Statistics supplies a pipeline: write a model, estimate unknown parts of the model, quantify whether data contradict a baseline claim, then choose an action whose expected loss is acceptable.

## Tier 1: four different questions

The opening challenge collapsed **four distinct tasks** into one number. Each question requires its own tool:

- A **parameter estimate** answers “what unknown value best fits the data?”
- **Uncertainty quantification** or a **hypothesis test** answers “would this data be surprising under a baseline claim?”
- A **prediction** answers “what might happen in the next observation?”
- A **decision** answers “which action has the best loss/utility tradeoff?”

Do not let one number answer four questions.

| Object             | Question it answers                                   | Campus version                                                    |
|--------------------|-------------------------------------------------------|-------------------------------------------------------------------|
| Parameter estimate | What value of an unknown quantity fits the data?      | $\hat{p} = 0.4$ from 8 rainy days out of 20.                      |
| Uncertainty / test | Would this data be surprising under a baseline claim? | Is 8 out of 20 surprising if ordinary event days have $p = 0.3$ ? |
| Prediction         | What might happen tomorrow or next event day?         | What is $P(\text{rain tomorrow} \mid \text{past data})$ ?         |
| Decision           | Which action has the best loss/utility tradeoff?      | Move indoors, stay outdoors, or wait for more information?        |

## Tier 2: statistical decision problem

A compact formal version is:

**Definition 1:** A **statistical model** is a **family of probability distributions**  $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$  for observable data  $X$ . Here  $\theta$  is an unknown parameter (or vector of parameters) and  $\Theta$  is the parameter space — the set of all possible parameter values. A **procedure** is a function of the data. It may return an estimate  $\hat{\theta}$ , a test decision  $\varphi(X) \in \{0, 1\}$ , a prediction, or an action  $a(X)$ .

In the running example,  $\theta = p$  is the unknown rain probability,  $X = (X_1, \dots, X_n)$  records rainy or dry event days (each  $X_i \in \{0, 1\}$ ),  $\varphi(X)$  may reject the claim  $p \leq 0.3$ , and  $a(X)$  may be “move indoors” or “stay outdoors.”

For estimation, the target is usually a **parameter**  $\theta$ . For testing, the target is a claim such as  $H_0 : \theta \in \Theta_0$ . For decision making, the target is an action  $a$  and a loss. The loss may compare  $a$  with a parameter, or with a future outcome  $Y$  that the team will experience.

## Tier 3: compare pooled and conditional models

Suppose the 20 recorded events contain 15 summer days with 3 rainy days and 5 winter days with 5 rainy days. The pooled estimate is

$$\hat{p}_{\text{pooled}} = \frac{8}{20} = 0.40.$$

For a future summer event, a model that conditions on season gives

$$\hat{p}_{\text{summer}} = \frac{3}{15} = 0.20.$$

Neither calculation is algebraically wrong. They estimate different parameters:  $P(\text{rain} \mid \text{recorded event day})$  and  $P(\text{rain} \mid \text{summer event day})$ . The second target matches a future summer event more closely, while the first mixes two populations with different rain rates.

The additional insight is that the model determines what the parameter means. A pooled Bernoulli model is useful only when the recorded days are representative of the future day. The summer estimate is still uncertain because it uses only 15 observations.

#### Capability Checkpoint: Inference pipeline

Work the numbered questions Q1–Q2 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 2. Parameter estimation: estimator, estimate, loss

The campus operations team has 20 historical event days: 8 had rain, 12 did not. Tomorrow’s outdoor event costs \$500 to move indoors, but \$5000 if equipment gets damaged by rain. Should they move indoors?

This decision needs three things: (1) an estimate of rain probability, (2) a **rule for making estimates from data**, and (3) a way to evaluate whether the rule is good enough.

### Tier 1: from data to decision

The **four-step inference pipeline**:

1. **Define the parameter:**  $p$  = probability of rain on an event day
2. **Propose an estimator:**  $\hat{p} = (\text{rainy days}) / (\text{total days})$
3. **Compute the estimate:**  $\hat{p} = \frac{8}{20} = 0.4$
4. **Make a decision:** If  $\hat{p} > \text{threshold}$ , move indoors

But consider **two candidate estimators**:

- **Estimator A:**  $\hat{p}_A = \bar{X}$  (sample proportion)
- **Estimator B:**  $\hat{p}_B = \frac{\bar{X} + 0.5}{2}$  (shrinks toward 0.5)

Both are rules. **Both map data to estimates**. For the observed data (8 rainy out of 20):

- Estimator A gives:  $\hat{p}_A = 0.4$
- Estimator B gives:  $\hat{p}_B = 0.45$

Which is better? **We need a loss function to answer this.** A loss function  $L(p, \hat{p})$  measures the cost of reporting estimate  $\hat{p}$  when the true probability is  $p$ . We then compare estimators by their **risk**: the expected loss averaged over all possible samples. If the true  $p = 0.3$ , estimator A has lower risk; if  $p = 0.5$ , estimator B might win. Tier 2 formalizes this comparison.

**Reference class matters.** The symbol  $p$  represents different probabilities depending on the conditioning: probability of rain given event day versus probability of rain given any day. Event days might cluster in certain seasons or correlate with weather patterns, changing the probability. Always specify the sample space.

### Wrong Path: Estimate/procedure confusion

“The estimator is 0.4” is imprecise. The **estimator is the rule**  $\hat{p} = \bar{X}$ . The estimate is 0.4 after the observed sample is plugged in. Before observing data, the estimator is a random variable; after observing data, the estimate is a fixed number.

### Pause and Think

**Question:** If we could repeat the sampling process and observe different event-day samples from the same underlying process, which of the following would change: (a) the true parameter  $p$ , (b) the estimator rule  $\hat{p} = \bar{X}$ , (c) the numerical estimate?

*Hint:* The true parameter  $p$  is a property of the process, not the sample. The estimator is the rule, not tied to one dataset.

**Answer:**

(a) The true  $p$  stays fixed—it’s a property of the weather process for event days. (b) The estimator rule  $\hat{p} = \bar{X}$  stays the same—it’s our chosen procedure. (c) The numerical estimate changes with each new sample because  $\bar{X}$  is random before we observe the data.

| Concept        | Symbol | Type          | Campus example                             |
|----------------|--------|---------------|--------------------------------------------|
| True parameter | $p$    | Fixed unknown | 0.35 (true rain probability on event days) |

| Concept   | Symbol              | Type            | Campus example                         |
|-----------|---------------------|-----------------|----------------------------------------|
| Estimator | $\hat{p} = \bar{X}$ | Random function | Rule: average the rain indicators      |
| Estimate  | 0.4                 | Number          | After observing 8 rainy days out of 20 |

## Tier 2: tools for evaluating estimators

We have two candidate estimators. To choose between them, we need to measure the cost of being wrong.

### Loss functions

**Definition 2:** For parameter estimation, a **loss function**  $L(\theta, a)$  measures the **cost of reporting numerical estimate**  $a$  when the true parameter is  $\theta$ .

The indoor/outdoor decision has discrete actions and a future rain outcome. Its operational loss is therefore not a numerical estimation-error loss  $L(p, a)$ . Section 12 defines that action loss and averages it using a predictive rain probability.

Common loss functions for parameter estimation:

- **Squared error:**  $(\theta - a)^2$  treats overestimation and underestimation symmetrically
- **Absolute error:**  $|\theta - a|$  also symmetric, but less sensitive to large errors
- **0–1 loss:**  $\mathbf{1}_{a \neq \theta}$  when both the target and reported action are discrete

### Expected loss and risk

A single loss  $L(p, \hat{p})$  depends on the unknown  $p$  and the random estimate  $\hat{p}$ . To evaluate an estimator, we **average over all possible samples**:

**Definition 3:** The **risk** of estimator  $\hat{\theta}$  under parameter  $\theta$  is

$$R(\theta, \hat{\theta}) = E[L(\theta, \hat{\theta}(X_1, \dots, X_n))],$$

where the expectation is over all possible samples from the model with parameter  $\theta$ .

For the campus example, if the true  $p = 0.3$  and we use estimator A ( $\hat{p}_A = \bar{X}$ ) with squared loss, the risk is

$$R(0.3, \hat{p}_A) = E[(0.3 - \bar{X})^2] = \text{Var}(\bar{X}) = \frac{p(1-p)}{n} = 0.3 \cdot \frac{0.7}{20} = 0.0105.$$

Different true  $p$  values give different risks. Estimator B (which shrinks toward 0.5) might have lower risk when  $p$  is near 0.5, but higher risk when  $p$  is near 0 or 1.

### Two frameworks for inference

Now we face a choice: how do we pick the “best” estimator when risk depends on the unknown  $p$ ?

**Definition 4:** In a **frequentist** estimation problem, the parameter  $\theta$  **is treated as fixed but unknown**. An estimator  $\hat{\theta} = g(X_1, \dots, X_n)$  is a function that maps data to a parameter estimate. It is evaluated by its sampling behavior under each possible  $\theta$ : if we repeatedly draw samples from the model with the same true  $\theta$ , how does the estimator perform on average?

Frequentist question: “What if we could repeat this experiment many times with the same true  $p$ ?” We evaluate risk  $R(p, \hat{p})$  for each possible  $p$ , and choose estimators that minimize worst-case risk, or are unbiased ( $E[\hat{p}] = p$ ), or have other desirable properties.

**Definition 5:** In a **Bayesian** estimation problem,  $\theta$  is modeled as uncertain with prior distribution  $P(\theta)$ . The prior encodes knowledge or assumptions before observing data. After data  $D$ , inference uses the posterior distribution

$$P(\theta | D) = \frac{P(\theta)P(D | \theta)}{\int P(\theta)P(D | \theta) d\theta}.$$

The denominator is a normalizing constant that ensures the posterior integrates to 1. Use a sum rather than an integral when the parameter space is discrete.

Bayesian question: “What if  $p$  itself is uncertain, and we have prior information?” For the campus case, historical climate data might suggest  $p \sim \text{Beta}(a, b)$ . After observing 8 rainy days out of 20, the posterior is  $p | \text{data} \sim \text{Beta}(a + 8, b + 12)$ . The Bayesian estimate might be the posterior mean, which balances prior information and observed data.

### Link between loss and estimator choice

Different loss functions favor different estimators:

| Context                     | Loss                         | Optimal Bayes rule   |
|-----------------------------|------------------------------|----------------------|
| Parameter estimate          | $(\theta - a)^2$             | posterior mean       |
| Parameter estimate          | $ \theta - a $               | posterior median     |
| Discrete parameter or state | $\mathbf{1}_{\theta \neq a}$ | posterior mode / MAP |

For Bernoulli data,  $\bar{X}$  is the MLE and is unbiased for  $p$ . Frequentist choice still requires a criterion such as risk; loss alone does not make the sample mean or sample median universally optimal for every parameter. Operational action costs are handled separately in Section 12.

### Tier 3: construct the target before the estimator

Suppose equipment is damaged only by heavy rain during the event’s 2–5 p.m. operating window. The decision-relevant target is

$$q = P(\text{heavy rain during 2--5 p.m.} \mid \text{planned event conditions}),$$

not the probability of any rain during an entire day. A low-MSE estimator of the all-day probability can still estimate the wrong quantity for this action.

#### Practice: Parameter-estimation design brief

State the action, parameter and time window, data coding, estimator, estimation loss, and one assumption needed to transfer the estimate to the planned event.

Define the event, time window, and conditioning information before choosing an estimator. Historical selection bias remains an important qualification.

#### Capability Checkpoint: Estimator, estimate, and loss

Work the numbered questions Q3–Q5 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

### 3. Maximum likelihood: make the observed data plausible

MLE asks a terse question: if each candidate parameter were true, how plausible would the observed data be? The estimate is the parameter that **makes the data most likely** under the model.

#### Tier 1: the rainy-day MLE

Suppose  $X_i \in \{0, 1\}$  (where 1 means rain and 0 means no rain) and  $X_i \mid p \sim \text{Bernoulli}(p)$  independently. This means each day's outcome is a **coin flip with success probability**  $p$ . If 8 of 20 days had rain, the MLE is the sample frequency:

$$\hat{p}_{\text{MLE}} = \frac{8}{20} = 0.4.$$

This is intuitive, but the likelihood derivation matters because the **same recipe works when the answer is not obvious**.

#### Tier 2: Bernoulli likelihood derivation

Let  $m = \sum_i X_i$  be the number of rainy days and  $n - m$  the number of dry days. For the observed sequence, the likelihood is

$$L(p) = P(D \mid p) = \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} = p^m (1-p)^{n-m}.$$

If we record only the count  $m$ , a factor  $\binom{n}{m}$  appears, but it **does not depend on**  $p$  and therefore does not affect the MLE.

The log-likelihood is

$$\ell(p) = \log L(p) = m \log p + (n - m) \log(1 - p).$$

For  $0 < p < 1$ ,

$$\frac{d\ell}{dp} = \frac{m}{p} - \frac{n - m}{1 - p}.$$

Setting the derivative to zero gives

$$m(1 - p) = (n - m)p \Rightarrow \hat{p}_{\text{MLE}} = \frac{m}{n}.$$

If  $m = 0$  or  $m = n$ , the derivative equation has no interior solution; the **maximum occurs at the boundary**  $p = 0$  or  $p = 1$ .

#### Pause and Think

**Question:** Why maximize the log-likelihood instead of the likelihood?

*Hint:* The logarithm is increasing, so it preserves the maximizer. Products become sums.

**Answer:**

log is monotone, so  $\arg \max_p L(p) = \arg \max_p \log L(p)$ . The log form is easier to differentiate and numerically more stable.

#### Tier 3: a boundary MLE still leaves nearby values plausible

Suppose no rain is observed in five event days. Tier 2 gives  $\hat{p}_{\text{MLE}} = 0$  because

$$L(p) = (1 - p)^5$$

is largest at the boundary  $p = 0$ . But compare  $p = 0$  with  $p = 0.1$ :

$$\frac{L(0.1)}{L(0)} = 0.9^5 \approx 0.590.$$

The value  $p = 0.1$  retains about 59% of the maximum likelihood. Thus the MLE ranks  $p = 0$  highest; [it does not prove that rain is impossible](#) or supply an uncertainty statement.

### Wrong Path: No rain means no risk

If the team observes no rain in five event days, the Bernoulli MLE is  $\hat{p}_{\text{MLE}} = 0$ . That does not mean rain is impossible. It means  $p = 0$  maximizes this small-sample likelihood under the assumed model. A decision that treats this estimate as certainty may be fragile.

The important qualification is sample size: the same boundary estimate after 500 dry days has much stronger likelihood support against moderate positive values of  $p$  than it does after five days.

### Capability Checkpoint: Likelihood and Bernoulli MLE

Work the numbered questions Q6–Q8 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 4. Bayesian estimation: prior + likelihood $\rightarrow$ posterior

The campus team may not be starting from zero. Historical climate records may suggest that event-day rain is usually near 25%. Bayesian estimation gives a formal way to **combine that prior information with the observed event days**.

### Tier 1: prior data and observed data both count

For rainy/not-rainy data, a convenient prior is  $p \sim \text{Beta}(a, b)$ . The Beta distribution is a probability distribution on the interval  $[0, 1]$  controlled by two parameters  $a > 0$  and  $b > 0$ . The parameters  $a$  and  $b$  behave like **prior pseudocounts for posterior means**: roughly,  $a$  represents previous rainy evidence and  $b$  represents previous dry evidence. For posterior modes, the -1 terms in the Beta density matter. After observing  $m$  rainy days out of  $n$ , the posterior is another Beta distribution:

$$p \mid D \sim \text{Beta}(a + m, b + n - m).$$

The posterior mean is

$$E[p \mid D] = \frac{a + m}{a + b + n},$$

which **shrinks the sample frequency** toward the prior mean.

### Tier 2: MAP and posterior mean

The Beta density has the form

$$f(p) \propto p^{a-1}(1-p)^{b-1}, \quad 0 < p < 1.$$

This probability density function is defined over  $p \in [0, 1]$  and peaks near  $\frac{a-1}{a+b-2}$  when  $a, b > 1$ .

Multiplying prior and likelihood:

$$P(p \mid D) \propto p^{a-1}(1-p)^{b-1} \cdot p^m(1-p)^{n-m} = p^{a+m-1}(1-p)^{b+n-m-1}.$$

Thus

$$p \mid D \sim \text{Beta}(a + m, b + n - m).$$

For interior modes when  $a + m > 1$  and  $b + n - m > 1$ ,

$$\hat{p}_{\text{MAP}} = \frac{a + m - 1}{a + b + n - 2}.$$

The **MAP** (maximum a posteriori) estimate is the mode of the posterior distribution — the parameter value with highest posterior density.

But MAP is not the only way to extract a point estimate from a posterior. The **Bayes estimator** is the decision rule that minimizes expected loss under the **posterior distribution**. Different loss functions produce different Bayes estimators:

**Theorem 1:** The Bayes estimator under squared-error loss is the posterior mean:

$$\hat{p}_B = E[p \mid D] = \frac{a + m}{a + b + n}.$$

*Proof:* The Bayes estimator minimizes posterior expected loss:

$$a^* = \arg \min_a E_{p \mid D}[L(p, a)] = \arg \min_a E_{p \mid D}[(p - a)^2].$$

Expanding the squared error and taking the derivative with respect to  $a$ , the minimizer is  $a^* = E[p|D]$ , the posterior mean.  $\square$

The posterior mean can be written as a **weighted average of prior mean** and sample proportion:

$$\hat{p}_B = \left( \frac{a+b}{a+b+n} \right) \cdot \left( \frac{a}{a+b} \right) + \left( \frac{n}{a+b+n} \right) \cdot \left( \frac{m}{n} \right).$$

This decomposition reveals how much the prior ( $\frac{a}{a+b}$ ) versus the data ( $\frac{m}{n}$ ) influences the final estimate. The weight  $a+b$  is the prior's effective sample size: if  $a+b=80$  and  $n=20$ , the **prior contributes four times as much weight as the data**.

### Tier 3: measure prior sensitivity

For the same 8 rainy days out of 20, compare three posterior means:

| Prior        | Prior size | Posterior    | Posterior mean                |
|--------------|------------|--------------|-------------------------------|
| Beta(1, 1)   | 2          | Beta(9, 13)  | $\frac{9}{22} \approx 0.409$  |
| Beta(3, 7)   | 10         | Beta(11, 19) | $\frac{11}{30} \approx 0.367$ |
| Beta(20, 60) | 80         | Beta(28, 72) | $\frac{28}{100} = 0.280$      |

The calculation makes the source of the difference visible: a larger value of  $a+b$  gives the prior more weight relative to the 20 observations. A Beta(1,1) prior is uniform on the  $p$  scale, but that does not make it neutral under every reparameterization.

#### Wrong Path: Uniform prior arithmetic trap

A uniform prior on  $p$  is Beta(1, 1). After  $m$  rainy days out of  $n$ , the posterior is Beta( $m+1$ ,  $n-m+1$ ), the posterior mean is  $\frac{m+1}{n+2}$ , and the interior MAP is  $\frac{m}{n}$  when  $0 < m < n$ . Do not add 1 to the MAP numerator by confusing posterior mean and posterior mode.

#### Practice: Explain the sensitivity

State which prior has the greatest influence and what external evidence would justify treating its 80 pseudocounts as relevant to the planned event. If the prior is tuned using the same data, state that the procedure has changed and must account for that reuse of data.

Prior sensitivity is informative only when the **compared priors are plausible before the current analysis**; arbitrary extreme priors do not measure a useful uncertainty.

#### Capability Checkpoint: Bayesian parameter estimation

Work the numbered questions Q9–Q12 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 5. MLE, MAP, and Bayes estimates are different summaries

After Bayesian updating, the **posterior is a distribution**. A point estimate is a summary of that distribution, and different summaries serve different losses. The question is not “which formula is philosophically best?” The question is “when do these estimates diverge, and when does that divergence change your decision?”

### Tier 1: one posterior, three numbers

For the same rainy-day data (8 rainy out of 20) under a Beta(3, 7) prior:

- **MLE** summarizes the **likelihood alone**:  $\hat{p}_{\text{MLE}} = \frac{8}{20} = 0.40$
- **MAP** summarizes the **posterior by its mode**:  $\hat{p}_{\text{MAP}} = \frac{3+8-1}{3+7+20-2} = \frac{10}{28} \approx 0.357$
- **Bayes posterior mean** (squared-error loss):  $\hat{p}_B = \frac{3+8}{3+7+20} = \frac{11}{30} \approx 0.367$

The estimates differ by about 4 percentage points. **Whether that matters depends on your decision threshold**.

### Tier 2: when estimates diverge and why it matters

Three regimes of data vs. prior strength

| Scenario                                                        | MLE                     | MAP                                      | Post. mean                           |
|-----------------------------------------------------------------|-------------------------|------------------------------------------|--------------------------------------|
| Small data, strong prior<br>$n = 5, m = 2, \text{Beta}(10, 10)$ | $\frac{2}{5} = 0.40$    | $\frac{10+2-1}{20+5-2} \approx 0.478$    | $\frac{10+2}{20+5} = 0.48$           |
| Medium data, weak prior<br>$n = 20, m = 8, \text{Beta}(3, 7)$   | $\frac{8}{20} = 0.40$   | $\frac{3+8-1}{3+7+20-2} \approx 0.357$   | $\frac{3+8}{3+7+20} \approx 0.367$   |
| Large data, any prior<br>$n = 200, m = 80, \text{Beta}(3, 7)$   | $\frac{80}{200} = 0.40$ | $\frac{3+80-1}{3+7+200-2} \approx 0.394$ | $\frac{3+80}{3+7+200} \approx 0.395$ |

In the small-data regime, the prior’s effective sample size ( $a + b = 20$ ) dominates the data ( $n = 5$ ), pulling both MAP and posterior mean toward the prior mean  $\frac{10}{20} = 0.50$ . With only 2 successes out of 5 (sample proportion 0.4), the posterior mean is  $\frac{10+2}{20+5} = 0.48$ , very close to the prior mean. The **prior overwhelms the sparse data**.

In the medium-data regime, data weight  $n = 20$  and prior weight  $a + b = 10$  are comparable. MLE ignores the prior; MAP and posterior mean incorporate it and shift toward the prior mean 0.30.

In the large-data regime, data weight  $n = 200$  swamps the prior weight  $a + b = 10$ . **All three estimates converge** near the sample proportion 0.40.

### Mathematical explanation of convergence

The posterior mean decomposes as:

$$\hat{p}_B = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{m}{n}.$$

As  $n \rightarrow \infty$ , the data weight  $\frac{n}{a+b+n} \rightarrow 1$ , so the posterior mean converges to  $\frac{m}{n}$ . The MAP also concentrates near the likelihood peak. For large  $n$ , all three estimates agree.

For small  $n$ , if the prior’s effective sample size  $a + b$  is large relative to  $n$ , the prior dominates; if  $a + b$  is small, MLE and Bayes estimates are close.

### Operational consequence: threshold decisions

Campus decision: move indoors if  $p > 0.35$ .

With 8/20 rainy days under Beta(3,7) prior:

- MLE = 0.40 → move indoors
- MAP  $\approx 0.357$  → move indoors (barely crosses threshold)
- Posterior mean  $\approx 0.367$  → move indoors

All three agree on the action.

But change the prior to Beta(15, 35) (same prior mean 0.3, but stronger effective sample size 50):

- MLE = 0.40 → move indoors
- MAP =  $\frac{15+8-1}{15+35+20-2} = \frac{22}{68} \approx 0.324$  → stay outdoors
- Posterior mean =  $\frac{15+8}{15+35+20} = \frac{23}{70} \approx 0.329$  → stay outdoors

**A strong prior flips the decision.** The **prior's 50 pseudocounts** (reflecting, say, decades of historical climate data) outweigh the 20 observed event days, pulling the estimate below the 0.35 threshold.

This shows why estimate choice matters: it's not just three different numbers, it's potentially three different actions.

### Loss function as the tiebreaker

When estimates diverge, the loss function determines which to use:

| Loss function                    | Optimal Bayes estimator    | When to use                                        |
|----------------------------------|----------------------------|----------------------------------------------------|
| Squared error $(p - a)^2$        | Posterior mean $E[p D]$    | Penalizes large errors heavily; standard default   |
| Absolute error $ p - a $         | Posterior median           | Robust to outliers and skew                        |
| 0–1 loss $\mathbf{1}_{p \neq a}$ | MAP (in discrete settings) | Only care about exact match or most probable value |

MLE is a likelihood summary rather than the optimizer of one of these posterior expected losses.

For the campus case with Beta(11, 19) posterior:

- Posterior mean:  $\frac{11}{30} \approx 0.367$
- Posterior median: solve  $\int_0^{\text{med}} \text{Beta}(p; 11, 19) dp = 0.5$  numerically → median  $\approx 0.363$
- MAP:  $\frac{11-1}{11+19-2} = \frac{10}{28} \approx 0.357$

The median sits between MAP and mean because the Beta(11,19) distribution is slightly right-skewed. Under absolute-error loss, use the median; under squared-error loss, use the mean.

### Asymptotic equivalence

**Theorem 2:** (Bernstein–von Mises, informal) Let  $I_1(\theta_0)$  be the Fisher information in one observation at the true parameter  $\theta_0$ . Under regularity conditions, as  $n \rightarrow \infty$  the posterior is approximately normal, centered near the MLE, with variance  $[nI_1(\theta_0)]^{-1}$ . Consequently, MAP, MLE, and posterior mean converge to the same value.

**Interpretation:** For large samples, the choice between MLE, MAP, and posterior mean doesn't matter much—they all converge. For small samples, they can diverge significantly, and the choice matters for decisions.

### Tier 3: quantify what a point estimate omits

Both samples 2/5 and 200/500 have Bernoulli MLE  $\hat{p} = 0.4$ . Under a Beta(1, 1) prior, however, the posteriors are Beta(3, 4) and Beta(201, 301). For a Beta( $u, v$ ) distribution,

$$\text{Var}(p \mid D) = \frac{uv}{(u+v)^2(u+v+1)}.$$

Therefore the posterior standard deviations are approximately

$$\sqrt{\frac{12}{7^2 \cdot 8}} \approx 0.175$$

and

$$\sqrt{\frac{201 \cdot 301}{502^2 \cdot 503}} \approx 0.022.$$

The same MLE can therefore accompany very different uncertainty. A point estimate is adequate only when the action or report does not need the missing spread or tail probabilities.

**Remark.** For continuous parameters, MAP is a density mode, not a probability mass. It can also change under reparameterization, so treat it as a useful summary rather than an invariant statement about “the most probable exact value.”

#### Capability Checkpoint: MLE, MAP, Bayes estimator

Work the numbered questions Q13–Q15 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 6. Gaussian MLE: the sample mean is not magic

Everyone “knows” that to estimate a mean, you compute the sample average. But why? Is it because it’s simple, or because it’s optimal in some sense? This section shows that the **sample mean emerges from maximum likelihood estimation** for Gaussian data—it’s not just convenient, it’s the parameter value that **makes your observed data most plausible** under the model.

### Tier 1: two questions about forecast scores

Let  $R$  be the number of numerical forecast scores, written  $X_1, \dots, X_R$ . For  $R = 20$ , suppose  $X_1 = 72, X_2 = 81, X_3 = 68, \dots, X_{20} = 77$  with  $\bar{X} = 75$ .

**Question 1:** If scores are  $\text{Normal}(\mu, \sigma^2 = 100)$ , which  $\mu$  value makes these 20 observations most likely?

**Setup:** We know the spread ( $\sigma^2 = 100$ ) but not the center ( $\mu$ ).

**Answer:**  $\hat{\mu}_{\text{MLE}} = 75$  (the sample mean).

**Question 2:** If we don’t know  $\sigma^2$  either, what are the best-fitting values of both  $\mu$  and  $\sigma^2$ ?

**Setup:** Estimate both parameters jointly.

**Answer:**  $\hat{\mu}_{\text{MLE}} = 75, \hat{\sigma}_{\text{MLE}}^2 = \left(\frac{1}{20}\right) \sum_{i=1}^{20} (X_i - 75)^2$ .

**Why this matters:** The **sample mean isn’t arbitrary—it’s the MLE**. Under Gaussian assumptions, it’s the “most believable” center given the data.

### Tier 2: the mathematical justification

#### Problem 1: Estimate $\mu$ when $\sigma^2$ is known

For i.i.d. data  $X_1, \dots, X_R \simeq \text{Normal}(\mu, \sigma^2)$  with known  $\sigma^2$ , the likelihood is:

$$L(\mu) = \prod_{i=1}^R \left( \frac{1}{\sqrt{2\pi}\sigma} \right) \exp \left( -\frac{(X_i - \mu)^2}{2\sigma^2} \right).$$

Taking logs:

$$\ell(\mu) = \text{constant} - \left( \frac{1}{2\sigma^2} \right) \sum_{i=1}^R (X_i - \mu)^2.$$

Maximizing  $\ell(\mu)$  is equivalent to **minimizing the sum of squared deviations**:

$$\sum_{i=1}^R (X_i - \mu)^2.$$

Differentiating with respect to  $\mu$ :

$$\frac{d}{d\mu} \sum_{i=1}^R (X_i - \mu)^2 = -2 \sum_{i=1}^R (X_i - \mu).$$

Setting to zero:  $\sum_{i=1}^R (X_i - \mu) = 0$ , which gives

$$\hat{\mu}_{\text{MLE}} = \bar{X} = \left( \frac{1}{R} \right) \sum_{i=1}^R X_i.$$

For continuous data, likelihood is built from **density values, not probabilities** of exact observed points.

**Remark. MLE for Gaussian mean is equivalent to least-squares estimation.** Maximizing likelihood under Gaussian noise is the same as minimizing squared error between data and model prediction.

**Problem 2: Estimate both  $\mu$  and  $\sigma^2$** 

If both  $\mu$  and  $\sigma^2$  are unknown, the joint log-likelihood is:

$$\ell(\mu, \sigma^2) = \text{constant} - \left(\frac{R}{2}\right) \log \sigma^2 - \left(\frac{1}{2\sigma^2}\right) \sum_{i=1}^R (X_i - \mu)^2.$$

Take partial derivatives:

$$\frac{\partial \ell}{\partial \mu} = \left(\frac{1}{\sigma^2}\right) \sum_{i=1}^R (X_i - \mu),$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{R}{2\sigma^2} + \left(\frac{1}{2\sigma^4}\right) \sum_{i=1}^R (X_i - \mu)^2.$$

Setting both to zero and solving:

From the first equation:  $\hat{\mu}_{\text{MLE}} = \bar{X}$ .

Substituting into the second equation:

$$\hat{\sigma}_{\text{MLE}}^2 = \left(\frac{1}{R}\right) \sum_{i=1}^R (X_i - \bar{X})^2.$$

The joint MLE gives the same mean estimator as the known-variance case, but now **also estimates the spread from the data**.

Notice  $\hat{\sigma}_{\text{MLE}}^2$  uses divisor  $R$ , not  $R - 1$ . Section 7 explains why this MLE is biased and when to use the unbiased version  $\left(\frac{1}{R-1}\right) \sum (X_i - \bar{X})^2$ .

**Tier 3: when the score equations are coupled**

The Gaussian calculation in Tier 2 works because the score equations have closed-form solutions. A mixture model shows what changes when each observation has an unknown component label.

**Example: Why mixture models have no closed form**

Consider a mixture of two Gaussians: each observation comes from either  $\text{Normal}(\mu_1, \sigma_1^2)$  with probability  $w$  or  $\text{Normal}(\mu_2, \sigma_2^2)$  with probability  $1 - w$ . The likelihood for a single observation is

$$L(X_i | \mu_1, \sigma_1^2, \mu_2, \sigma_2^2, w) = w f_1(X_i; \mu_1, \sigma_1^2) + (1 - w) f_2(X_i; \mu_2, \sigma_2^2),$$

where  $f_1, f_2$  are Gaussian densities. The log-likelihood for all data involves:

$$\log \prod_{i=1}^R [w f_1(X_i) + (1 - w) f_2(X_i)] = \sum_{i=1}^R \log [w f_1(X_i) + (1 - w) f_2(X_i)].$$

The derivatives exist, but each score equation contains weights that depend on all the unknown parameters. The system generally has no closed-form solution and the log-likelihood can have multiple local maxima. Iterative methods such as expectation-maximization are therefore used. The MLE principle is unchanged; only the computation is harder.

**Capability Checkpoint: Gaussian MLE**

Work the numbered questions Q16–Q18 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 7. Bias, variance, MSE, and consistency

An estimator is a procedure. We **judge procedures before seeing the data** by asking how they behave over repeated samples from the model.

### Tier 1: unbiased is not the whole story

Three questions matter:

- **Bias:** Does the estimator hit the truth on average?
- **Variance:** How much does it fluctuate from sample to sample?
- **MSE:** How large is its squared error on average?

An estimator can be **biased but useful**, unbiased but noisy, or consistent even if it has finite-sample bias.

### Tier 2: definitions and decomposition

**Definition 6:** The **bias** of  $\hat{\theta}$  at parameter value  $\theta$  is

$$\text{Bias}_{\theta}(\hat{\theta}) = E_{\theta}[\hat{\theta}] - \theta.$$

An estimator is **unbiased** if its expected value equals the true parameter for all possible  $\theta$ .

The **mean squared error** is

$$\text{MSE}_{\theta}(\hat{\theta}) = E_{\theta}[(\hat{\theta} - \theta)^2].$$

MSE measures **average squared distance** from the estimator to the true parameter over repeated samples.

The core identity is

$$\text{MSE}_{\theta}(\hat{\theta}) = \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}_{\theta}(\hat{\theta})^2.$$

**Definition 7:** A sequence of estimators  $\hat{\theta}_n$  is **consistent** for  $\theta$  if  $\hat{\theta}_n \rightarrow \theta$  in probability as  $n \rightarrow \infty$ . This means the estimator **gets arbitrarily close to the true parameter** as the sample size grows.

For Gaussian data,  $\hat{\mu}_{\text{MLE}} = \bar{X}$  is unbiased:

$$E[\bar{X}] = \mu.$$

But the **MLE variance estimator is biased**:

$$E\left[\left(\frac{1}{R}\right) \sum_{i=1}^R (X_i - \bar{X})^2\right] = \left(\frac{R-1}{R}\right) \sigma^2.$$

The unbiased sample variance is

$$S^2 = \left(\frac{1}{R-1}\right) \sum_{i=1}^R (X_i - \bar{X})^2.$$

### Tier 3: choose a criterion before a denominator

The estimators  $\hat{\sigma}_{\text{MLE}}^2$  and  $S^2$  **optimize different criteria**. The first maximizes the Gaussian likelihood; the second is unbiased for  $\sigma^2$ . Neither property by itself proves lower MSE or better prediction.

#### Wrong Path: Unbiased means always better

The estimator with denominator  $R - 1$  is unbiased for  $\sigma^2$ . The estimator with denominator  $R$  is the Gaussian MLE. **Which is better depends on the criterion**: unbiasedness, MSE, likelihood fit, downstream prediction, or compatibility with a larger model.

#### Pause and Think

**Question:** Why should your intuition expect the MLE variance with denominator  $R$  to underestimate  $\sigma^2$ ?

*Hint:* The deviations are measured from  $\bar{X}$ , which was chosen from the same data to make deviations small.

**Answer:**

Estimating  $\mu$  from the same sample uses one degree of freedom. The fitted mean chases the sample, so residuals around it are too small on average compared with residuals around the true  $\mu$ .

#### Practice: Variance-estimator policy note

Choose between  $\hat{\sigma}_{\text{MLE}}^2$  and  $S^2$  for one task: fitting a Gaussian likelihood, following a reporting convention that requests an unbiased estimate, or minimizing prediction risk. State the criterion you are using; the third task requires a risk calculation rather than a rule of thumb.

For  $R = 5$  and  $S^2 = 10$ , the MLE is  $\hat{\sigma}_{\text{MLE}}^2 = \left(\frac{4}{5}\right)10 = 8$ . This numerical difference is the consequence of the **selected criterion**, not evidence that one formula is always correct.

#### Capability Checkpoint: Estimator quality

Work the numbered questions Q19–Q22 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 8. Hypothesis testing: from estimate to claim

The MLE gives  $\hat{p} = \frac{8}{20} = 0.4$ . Campus policy says “normal event days have rain risk at most 30%.” Should the team treat tomorrow’s event as “high risk” and change procedures? The **estimate alone doesn’t answer this**—it could be 0.4 by chance even if the true  $p = 0.3$ . Testing asks: **how surprising is this data under the baseline claim?**

### Tier 1: three questions that estimation alone can’t answer

Suppose you observed 8/20 rainy days two years ago ( $\hat{p} = 0.4$ ), then 7/20 last year ( $\hat{p} = 0.35$ ), then 9/20 this year ( $\hat{p} = 0.45$ ). All three estimates exceed 0.3, but is this pattern strong enough evidence that the true risk exceeds the policy baseline? Or is it consistent with natural year-to-year variation around  $p = 0.3$ ?

Estimation and testing answer different questions:

| Approach   | Question answered                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------|
| Estimation | “What is the best guess for $p$ ?” $\rightarrow \hat{p} = 0.4$                                      |
| Testing    | “Are the data surprising under baseline $p = 0.3$ ?” $\rightarrow$ Not surprising (p-value = 0.228) |

Three questions that estimation alone can’t answer:

1. **Is the observed  $\hat{p}$  materially different from the baseline?** (Could be sampling noise)
2. **If we acted on every estimate above 0.3, how often would we false-alarm?** (Type I error control)
3. **How often would the test detect a relevant alternative?** (Power at a specified value of  $p$ )

Testing provides a **formal framework** to answer these questions by quantifying surprise under a baseline claim.

### Tier 2: the testing framework with worked example

#### Step 1: Set up hypotheses

$$H_0 : p \leq 0.3 \quad (\text{baseline claim: normal risk}),$$

$$H_1 : p > 0.3 \quad (\text{elevated risk}).$$

The null  $H_0$  is the baseline, status-quo, or protected claim. We **provisionally keep**  $H_0$  unless the sample gives sufficiently strong evidence against it.

#### Step 2: Choose significance level

$\alpha = 0.05$  means we’re willing to falsely reject  $H_0$  at most 5% of the time when it’s actually true (Type I error rate).

#### Step 3: Choose test statistic

$$M = \sum_{i=1}^n X_i = \text{number of rainy days out of } n = 20.$$

Under  $H_0$ , if the true  $p = 0.3$  (worst case at boundary), then  $M \simeq \text{Binomial}(20, 0.3)$ .

#### Step 4: Find rejection region

We reject  $H_0$  for large  $M$  (one-sided test). Find cutoff  $c$  such that:

$$P(M \geq c \mid p = 0.3) \leq \alpha = 0.05.$$

Computing cumulative probabilities:

- $P(M \geq 9 \mid p = 0.3) = 0.113 > 0.05$

one-sided upper-tail test, the rejection probability  $P_p(M \geq c)$  increases with  $p$ . Among all null values  $p \leq 0.3$ , the largest false-alarm probability therefore occurs at the boundary  $p = 0.3$ . If Type I error is controlled there, it is controlled for every smaller  $p$  as well. The test is valid for the composite null, though it may still have low power against alternatives such as  $p = 0.4$ .

- $P(M \geq 10 \mid p = 0.3) = 0.048 \leq 0.05 \checkmark$

**Rejection region:**  $M \geq 10$

**Step 5: Apply to observed data**

Observed  $M = 8 < 10$ , so we **do not reject**  $H_0$ .

**Conclusion:** The **data are consistent with the baseline claim**  $p \leq 0.3$ . This doesn't prove the baseline is true—it means the evidence isn't strong enough to reject it.

**Step 6: Key concepts in context**

**Definition 8:** A test is a rule  $\varphi(X)$  where  $\varphi(X) = 1$  means reject  $H_0$  and  $\varphi(X) = 0$  means do not reject. The **significance level**  $\alpha$  is the maximum probability of a Type I error—rejecting  $H_0$  when it is true:

$$\sup_{\theta \in \Theta_0} P_\theta(\varphi(X) = 1) \leq \alpha.$$

**Definition 9:** A **Type I error** rejects  $H_0$  when  $H_0$  is true (false alarm).

A **Type II error** fails to reject  $H_0$  when the alternative  $H_1$  is true (missed detection).

The **power** at **parameter value**  $\theta$  is  $P_\theta(\varphi(X) = 1)$ , the probability of correctly rejecting when the alternative is true.

For our test (reject if  $M \geq 10$ ):

- **Type I error rate:**  $P(M \geq 10 \mid p = 0.3) = 0.048 \leq 0.05 \checkmark$
- **Type II error at  $p = 0.5$ :**  $P(M < 10 \mid p = 0.5) = 1 - 0.588 = 0.412$
- **Power at  $p = 0.5$ :**  $P(M \geq 10 \mid p = 0.5) = 0.588$

**Step 7: Power curve**

The test's ability to detect elevated risk depends on how much  $p$  exceeds 0.3:

| True $p$ | Power = $P(M \geq 10 \mid p)$ | Interpretation              |
|----------|-------------------------------|-----------------------------|
| 0.3      | $0.048 \leq \alpha$           | Type I error controlled     |
| 0.4      | 0.245                         | Only 24.5% chance to detect |
| 0.5      | 0.588                         | 58.8% chance to detect      |
| 0.6      | 0.873                         | 87.3% chance to detect      |

With  $n = 20$ , the test has low power for moderate increases ( $p = 0.4$ ). Even if the true risk is 50% (much higher than baseline), we'd only detect it about 59% of the time.

**Tier 3: compare tests with the same significance level**

Keep  $H_0 : p \leq 0.3$ ,  $H_1 : p > 0.3$ , and  $\alpha = 0.05$ . For each sample size, choose the smallest cutoff  $c$  satisfying  $P_{p=0.3}(M \geq c) \leq 0.05$ .

| $n$ | Reject when | Type I rate at $p = 0.3$ | Power at $p = 0.5$ |
|-----|-------------|--------------------------|--------------------|
| 20  | $M \geq 10$ | 0.048                    | 0.588              |
| 50  | $M \geq 21$ | 0.0478                   | 0.899              |

Both tests control the Type I error near 5%, but the larger sample detects  $p = 0.5$  much more often. With  $n = 20$ , a non-rejection is compatible with an alternative that this test misses 41.2% of the time. It therefore means insufficient evidence to reject, not acceptance of  $H_0$ .

Power must always name the alternative value: a test can have high power at  $p = 0.6$  and low power at  $p = 0.35$ .

#### Capability Checkpoint: Hypothesis-testing basics

Work the numbered questions Q23–Q26 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 9. p-values as calibrated surprise

Section 8 used a fixed cutoff: reject if  $M \geq 10$ , do not reject otherwise. This **binary decision loses information**. What if  $M = 9$  (just below cutoff)? What if  $M = 11$  (just above)? The p-value gives a **graded summary of surprise** under  $H_0$ . For this binomial test it can take only a discrete set of values.

### Tier 1: from a binary decision to graded evidence

Section 8's fixed rejection region:

- $M \geq 10$ : reject  $H_0 : p \leq 0.3$  at level  $\alpha = 0.05$
- $M < 10$ : do not reject

But this treats  $M = 9$  and  $M = 6$  identically (both “not rejected”), and  $M = 10$  and  $M = 15$  identically (both “rejected”). The **p-value restores the information lost by binary thresholding**.

**p-value comparison for rainy-day data ( $n = 20$ ,  $H_0 : p = 0.3$ ):**

| Observed $M$ | p-value | Reject at $\alpha = 0.05$ ? | Interpretation                    |
|--------------|---------|-----------------------------|-----------------------------------|
| 6            | 0.584   | No                          | Very consistent with $H_0$        |
| 7            | 0.392   | No                          | Moderate consistency              |
| 8            | 0.228   | No                          | Somewhat surprising, but not rare |
| 9            | 0.113   | No                          | Getting unusual under $H_0$       |
| 10           | 0.048   | Yes                         | Barely crosses threshold          |
| 11           | 0.017   | Yes                         | Quite rare under $H_0$            |

**Key observations:**

1. The p-value decreases in steps as the integer count  $M$  increases.
2. The 0.05 threshold is a convention, not a natural boundary.  $M = 9$  ( $p=0.113$ ) and  $M = 10$  ( $p=0.048$ ) are not fundamentally different in evidence strength.
3. Both  $M = 8$  and  $M = 9$  lead to “do not reject,” but their p-values (0.228 vs 0.113) reveal that  $M = 9$  is closer to the boundary of surprise.

### Tier 2: definition, computation, and testing frameworks

**Definition 10:** For a test statistic  $T(X)$  where larger values are more incompatible with  $H_0$ , the **p-value** for a composite null set  $\Theta_0$  is

$$\text{p-value} = \sup_{\theta \in \Theta_0} P_{\theta}(T(X^*) \geq T(x_{\text{obs}})),$$

where  $X^*$  is a fresh sample generated under parameter  $\theta$ . In this binomial upper-tail test, monotonicity makes the supremum over  $p \leq 0.3$  occur at the boundary  $p = 0.3$ .

The p-value for  $M = 8$  is computed as  $P_{p=0.3}(M \geq 8) = \sum_{k=8}^{20} 0.228$ .

**Computation for the rainy-day example:** Observed  $M = 8$  out of  $n = 20$ , testing  $H_0 : p = 0.3$  (one-sided upper-tail test for  $p > 0.3$ ).

Under  $H_0$ ,  $M \sim \text{Binomial}(20, 0.3)$ . The p-value is

$$\text{p-value} = P_{p=0.3}(M \geq 8) = \sum_{k=8}^{20} \binom{20}{k} 0.3^k 0.7^{20-k} = 0.228.$$

Since  $0.228 > 0.05$ , we do not reject  $H_0$  at the  $\alpha = 0.05$  level.

**Two philosophies of hypothesis testing:**

### 1. Neyman–Pearson framework (pre-specified $\alpha$ ):

- Fix the Type I error rate  $\alpha$  (e.g., 0.05) before seeing data.
- Choose a rejection region that controls  $P_{\text{Type I error}} \leq \alpha$ .
- Decision: reject if p-value  $\leq \alpha$ , otherwise do not reject.
- The p-value is compared to a threshold; its exact value matters only insofar as it crosses the line.

### 2. Fisherian framework (graded evidence):

- Report the p-value as a graded measure of evidence against  $H_0$ .
- No fixed  $\alpha$ ; readers judge the strength of evidence themselves.
- A p-value of 0.001 is stronger evidence than 0.04, even though both are “below 0.05.”

Both frameworks use the same p-value calculation. They differ in **whether you declare**  $\alpha$  upfront and make a binary decision, or report the p-value and let context determine action.

**Remark.** In simple-versus-simple testing (one specific null value  $p_0$  versus one specific alternative  $p_1$ ), the **Neyman–Pearson lemma** states that the likelihood-ratio test is most powerful among all tests with the same Type I error rate. For composite hypotheses, the lemma generalizes to uniformly most powerful tests when they exist.

## Tier 3: compare a p-value with a posterior probability

For  $M = 8$  out of 20, the one-sided p-value under  $H_0 : p = 0.3$  is 0.228. Now consider a different question with two simple models:

$$H_0 : p = 0.3, \quad H_1 : p = 0.5.$$

Assign prior probability  $\frac{1}{2}$  to each model. Because the common binomial coefficient cancels, Bayes’ rule gives

$$P(H_0 \mid M = 8) = \frac{0.3^8 0.7^{12}}{0.3^8 0.7^{12} + 0.5^{20}} \approx 0.488.$$

| Quantity              | Conditioning question                             | Value |
|-----------------------|---------------------------------------------------|-------|
| p-value               | How often is $M \geq 8$ under $p = 0.3$ ?         | 0.228 |
| Posterior probability | How probable is $H_0$ after comparing two models? | 0.488 |
| Observed effect       | How far is $\hat{p}$ from 0.3?                    | 0.10  |

These numbers differ because they answer different questions. The posterior probability also depends on the **chosen alternative and prior model probabilities**; the p-value does not supply either one. Neither number by itself determines the indoor/outdoor action.

### Capability Checkpoint: p-values and errors

Work the numbered questions Q27–Q30 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 10. Likelihood-ratio tests: a unified framework

Sections 8 and 9 used exact binomial calculations to test  $H_0 : p = 0.3$ . But what if the model is Gaussian, gamma, or a mixture? The **likelihood-ratio test** provides a general recipe: compare the **best fit under the null constraint** versus the best fit overall. If the constrained model fits much worse, the data push against the null.

### Tier 1: restricted fit versus unrestricted fit

**Question:** For the rainy-day data ( $m = 8$  out of  $n = 20$ ), how well does  $H_0 : p = 0.3$  explain the data compared to the unrestricted MLE?

**Step 1: Compute the best likelihood under  $H_0$ .** Under  $H_0 : p = 0.3$ , the likelihood is

$$L(p = 0.3) = \binom{20}{8} \cdot 0.3^8 \cdot 0.7^{12} = \binom{20}{8} \cdot 9.08 \times 10^{-7}.$$

**Step 2: Compute the best likelihood overall (unrestricted).** The MLE is  $\hat{p} = \frac{8}{20} = 0.4$ . The unrestricted likelihood is

$$L(\hat{p} = 0.4) = \binom{20}{8} \cdot 0.4^8 \cdot 0.6^{12} = \binom{20}{8} \cdot 1.43 \times 10^{-6}.$$

**Step 3: Compare the two.** The likelihood ratio is

$$\Lambda = \frac{L(p = 0.3)}{L(\hat{p} = 0.4)} = \frac{9.08 \times 10^{-7}}{1.43 \times 10^{-6}} \approx 0.637.$$

The null model achieves about 64% of the unrestricted likelihood. This ratio is a test statistic, **not a probability that the null** is true. To turn it into a p-value, we must specify an alternative and calibrate the statistic under the null.

### Tier 2: the likelihood-ratio statistic and Wilks' theorem

**Definition 11:** For null parameter set  $\Theta_0 \subset \Theta$ , the **generalized likelihood-ratio statistic** is

$$\Lambda(D) = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}.$$

The numerator is the best likelihood we can achieve while respecting the null constraint; the denominator is the best likelihood overall. Small  $\Lambda(D)$  indicates that **the null-constrained model fits much worse** than the unrestricted model.

For a simple null  $H_0 : p = p_0$  against unrestricted Bernoulli  $p$ , with  $0 < m < n$ ,

$$\Lambda(D) = \frac{p_0^m (1 - p_0)^{n-m}}{\left(\frac{m}{n}\right)^m \left(1 - \frac{m}{n}\right)^{n-m}}.$$

Often one works with the log-likelihood-ratio statistic:

$$-2 \log \Lambda(D) = 2 \left( \log L(\hat{\theta}) - \log L(\hat{\theta}_0) \right),$$

where  $\hat{\theta}$  is the unrestricted MLE and  $\hat{\theta}_0$  is the best fit under  $H_0$ .

For the rainy-day example:

$$-2 \log \Lambda = -2 \log(0.637) \approx 0.903.$$

**Wilks' theorem (informal statement):** Under regularity conditions, when  $H_0$  is true and  $n \rightarrow \infty$ , the statistic  $-2 \log \Lambda(D)$  converges in distribution to  $\chi_k^2$ , where  $k$  is the difference in the number of free parameters between the unrestricted and null models.

For the simple null  $H_0 : p = 0.3$  against the unrestricted two-sided alternative  $H_1 : p \neq 0.3$ ,  $k = 1$ . The asymptotic p-value is

$$P(\chi_1^2 \geq 0.903) \approx 0.342.$$

Do not compare this value directly with the one-sided upper-tail p-value 0.228 from Section 9. The two procedures use different alternatives and different rejection regions, so their difference is not an approximation error.

The factor of 2 in  $-2 \log \Lambda$  is conventional and makes the asymptotic distribution chi-square. Without the factor, the limit would be one-half of a chi-square variable.

### Why likelihood-ratio tests are powerful:

1. **Generality:** The same framework works for binomial, Gaussian, gamma, regression, and other models. You don't need to derive a new test statistic for each model family.
2. **Optimality:** For simple-versus-simple hypotheses (one specific  $H_0$  versus one specific  $H_1$ ), the likelihood-ratio test is the most powerful test at any given Type I error rate (Neyman–Pearson lemma).
3. **Asymptotic theory:** Wilks' theorem provides a general asymptotic distribution, avoiding the need for exact tail calculations when  $n$  is large and regularity conditions hold.

### Tier 3: calibrate the same statistic two ways

For each possible count  $k = 0, \dots, 20$ , compute the same two-sided likelihood-ratio statistic

$$D(k) = -2 \log \Lambda(k).$$

The observed count  $k = 8$  gives  $D(8) = 0.903$ . Under  $H_0 : p = 0.3$ , exact enumeration of the binomial counts gives

$$P_{p=0.3}(D(M) \geq 0.903) \approx 0.465.$$

Wilks' large-sample calibration gives  $P(\chi_1^2 \geq 0.903) \approx 0.342$ . These values now refer to the same statistic and two-sided alternative, so their difference is a genuine finite-sample calibration difference.

The chi-square reference can also fail in large samples when regularity is violated, for example by boundary parameters or non-identifiability. In those cases, exact enumeration, simulation under the null, or a problem-specific limiting distribution is needed.

#### Capability Checkpoint: Likelihood-ratio testing

Work the numbered questions Q31–Q33 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 11. Predictive distribution: tomorrow is not the parameter

Sections 8–10 tested whether the data are consistent with  $H_0 : p \leq 0.3$ . But the team doesn't experience the parameter  $p$  tomorrow—it **experiences rain or no rain**. Before making a decision (Section 12), we need a prediction: what is the probability of rain tomorrow?

### Tier 1: plug-in versus posterior predictive

A **frequentist plug-in prediction** uses the MLE as if it were the true parameter:

$$P(\text{rain tomorrow} \mid \text{past data}) \approx \hat{p} = \frac{8}{20} = 0.4.$$

This treats the MLE as certain and **ignores estimation uncertainty**.

A **Bayesian posterior predictive** **averages over parameter uncertainty**:

$$P(\text{rain tomorrow} \mid D) = \int P(\text{rain tomorrow} \mid p)P(p \mid D) dp.$$

For one future Bernoulli event day that is conditionally independent of the past given  $p$ , the posterior predictive **equals the posterior mean**:

$$P(\text{rain tomorrow} \mid D) = E[p \mid D] = \frac{a + m}{a + b + n}.$$

#### Concrete calculation for the rainy-day example:

- Data:  $m = 8$  rainy out of  $n = 20$  event days
- Prior:  $p \sim \text{Beta}(3, 7)$
- Posterior:  $p \mid D \sim \text{Beta}(11, 19)$

The two predictions are:

| Method               | Formula                              | Probability |
|----------------------|--------------------------------------|-------------|
| Plug-in              | $\hat{p} = \frac{8}{20}$             | 0.400       |
| Posterior predictive | $\frac{3+8}{3+7+20} = \frac{11}{30}$ | 0.367       |

The posterior predictive is lower here because it is a weighted average of the sample proportion 0.4 and the lower prior mean 0.3. Averaging over parameter uncertainty does not inherently lower a prediction; a prior mean above 0.4 would raise it.

### Tier 2: why integration appears

The Bayesian calculation uses the law of total probability:

$$P(X_{n+1} = 1 \mid D) = \int P(X_{n+1} = 1 \mid p, D)P(p \mid D) dp.$$

Under conditional independence given  $p$  (tomorrow's rain is independent of past days given the parameter),

$$P(X_{n+1} = 1 \mid p, D) = p,$$

so the predictive probability is

$$P(X_{n+1} = 1 \mid D) = \int p \cdot P(p \mid D) dp = E[p \mid D].$$

For our  $\text{Beta}(11, 19)$  posterior,

$$E[p \mid D] = \frac{11}{11 + 19} = \frac{11}{30} \approx 0.367.$$

The posterior predictive can be written as a weighted average:

$$\frac{a}{a+b} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{m}{n}$$

showing how the prior mean  $\left(\frac{a}{a+b}\right)$  and sample proportion  $\left(\frac{m}{n}\right)$  combine.

This is the probability we'll use in Section 12 to make the decision.

**Comparison table:**

| Quantity                | Value | Meaning        |
|-------------------------|-------|----------------|
| MLE $\hat{p}$           | 0.400 | Point estimate |
| Posterior mean $E[p D]$ | 0.367 | Predictive     |
| Prior mean              | 0.300 | Before data    |

The posterior predictive (0.367) lies between the MLE (0.400) and the prior mean (0.300), weighted by effective sample sizes.

### Tier 3: compare pooled and season-specific predictions

Return to the data split from Section 1: 3 rainy days among 15 summer events and 5 rainy days among 5 winter events. With a Beta(1, 1) prior, a pooled model predicts

$$P(\text{rain next} \mid D) = \frac{1 + 8}{2 + 20} = \frac{9}{22} \approx 0.409.$$

For a future summer event, a season-specific model predicts

$$P(\text{rain next summer event} \mid D_{\text{summer}}) = \frac{1 + 3}{2 + 15} = \frac{4}{17} \approx 0.235.$$

Both calculations integrate parameter uncertainty inside their chosen model. The **large difference comes from the model choice**: the pooled model treats summer and winter events as conditionally identically distributed, whereas the second model does not.

The season-specific estimate uses fewer observations, so it has greater parameter uncertainty. Posterior prediction handles that uncertainty, but it cannot decide by itself whether the seasonal split is the right model.

#### Capability Checkpoint: Predictive distribution

Work the numbered questions Q38–Q40 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 12. Decision theory: choose actions, not just estimates

Section 11 gave us two predictions for tomorrow's rain: plug-in (0.4) and posterior predictive (0.367). But the team doesn't need a probability—it needs a decision: move indoors or stay outdoors? Decision theory chooses the action that **minimizes expected loss**.

For estimation, loss may compare an action or reported value  $a$  with a parameter  $\theta$ . For operational decisions, loss usually compares an action  $a$  with the future outcome  $Y$  that the team will actually experience.

### Tier 1: action depends on loss

Let the team choose between:

- $a_{\text{out}}$ : stay outdoors;
- $a_{\text{in}}$ : move indoors.

Suppose moving indoors costs 10 units (in setup, inconvenience, or opportunity cost). Staying outdoors costs 0 if there is no rain and 60 if there is rain (equipment damage, cancellation, reputation). If the predicted probability of rain is  $q$ , the expected losses are

$$\text{risk}(a_{\text{in}}) = 10, \quad \text{risk}(a_{\text{out}}) = 60q.$$

Move indoors if  $60q > 10$ , i.e. if  $q > \frac{1}{6} \approx 0.167$ . The **decision threshold is not 50%**; it is determined by losses.

Using the predictions from Section 11:

| Method               | $q$   | $E[\text{loss} \mid \text{out}]$ | $E[\text{loss} \mid \text{in}]$ | Decision     |
|----------------------|-------|----------------------------------|---------------------------------|--------------|
| Posterior predictive | 0.367 | $60 \times 0.367 = 22.0$         | 10.0                            | Move indoors |
| Plug-in              | 0.400 | $60 \times 0.400 = 24.0$         | 10.0                            | Move indoors |

Both methods lead to “move indoors” because both predicted probabilities exceed the break-even threshold of 0.167. However, the margins differ:

- Posterior predictive: margin of  $22.0 - 10.0 = 12.0$  units
- Plug-in: margin of  $24.0 - 10.0 = 14.0$  units

The posterior-predictive margin is smaller here because the chosen prior pulls the predictive probability below the sample proportion. Parameter averaging alone does not imply a smaller margin.

### Tier 2: expected loss and Bayes action

**Definition 12:** Given posterior or predictive uncertainty, the **posterior expected loss** of action  $a$  is

$$\rho(a \mid D) = E[L(Y, a) \mid D].$$

A **Bayes action** **minimizes this expected loss**.

Loss functions explain common estimators:

| Loss     | Form                    | Optimal summary                 |
|----------|-------------------------|---------------------------------|
| 0–1 loss | $\mathbf{1}_{y \neq a}$ | mode / MAP in discrete settings |

| Loss          | Form                   | Optimal summary                     |
|---------------|------------------------|-------------------------------------|
| Squared loss  | $(y - a)^2$            | posterior mean                      |
| Absolute loss | $ y - a $              | posterior median                    |
| Cross-entropy | $-\sum_k y_k \log a_k$ | calibrated predictive probabilities |

For binary probabilistic prediction with true label  $y \in \{0, 1\}$  and predicted probability  $a$ , **cross-entropy loss** is

$$L(y, a) = -y \log a - (1 - y) \log(1 - a).$$

Cross-entropy measures how well a predicted probability distribution matches the true outcome. Its expected value differs from KL divergence by a term depending only on the true distribution, so minimizing expected cross-entropy encourages matching the predictive distribution.

### Tier 3: compare a test decision with a loss decision

The [hypothesis test and the action rule use different thresholds](#). The test rejects only when the data cross a rejection boundary chosen to control Type I error. The action rule moves indoors when the predictive rain probability exceeds the loss-based threshold  $\frac{10}{60} = \frac{1}{6}$ .

**Connecting the pipeline stages for the rainy-day example:**

| Stage                 | Question answered         | Result                         |
|-----------------------|---------------------------|--------------------------------|
| Estimate (Sec 3)      | What is $p$ ?             | MLE: $\hat{p} = 0.4$           |
| Uncertainty (Sec 4-5) | What uncertainty remains? | Posterior: Beta(11, 19)        |
| Test (Sec 8-9)        | Is $p > 0.3$ significant? | p-value = 0.228, do not reject |
| Predict (Sec 11)      | Will it rain tomorrow?    | $P(\text{rain}) = 0.367$       |
| Decide (Sec 12)       | Move indoors?             | Yes, $22.0 > 10.0$             |

Notice: The test says “do not reject  $H_0 : p \leq 0.3$ ” (p-value = 0.228 > 0.05), but the decision says “move indoors.” These are not contradictory. The test controls Type I error at 0.05. The decision minimizes expected loss given the break-even threshold of 0.167. Since our posterior predictive (0.367) exceeds 0.167, we move indoors—regardless of whether we rejected  $H_0$  at  $\alpha = 0.05$ .

#### Wrong Path: Reject therefore act

Rejecting  $H_0 : p \leq 0.3$  does not automatically mean move indoors. If moving indoors is extremely expensive, the optimal action may still be to stay outdoors. Conversely, failing to reject may still justify moving indoors if outdoor failure is catastrophic.

#### Practice: Loss-sensitive event decision

Choose numbers for four quantities: cost of moving indoors, cost of rain outdoors, predictive rain probability, and cost of waiting for one more forecast. Compute which available action has the smallest expected loss.

#### Capability Checkpoint: Decision theory and loss

Work the numbered questions Q34–Q37 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 13. Transfer: A/B testing applies the same pipeline

Sections 2–12 built the inference-to-decision pipeline using a single running example: campus rain days. This section applies the same pipeline to a different domain—A/B testing—to show that the **logic transfers**. We'll work through a concrete A/B test step-by-step, mapping each stage to the corresponding section from the rainy-day analysis.

### Tier 1: the A/B scenario and pipeline mapping

**Scenario:** A website tests two versions. Version A has 96 conversions out of 1000 visitors ( $\hat{p}_A = 0.096$ ). Version B has 112 conversions out of 1000 visitors ( $\hat{p}_B = 0.112$ ). Observed lift: 1.6 percentage points. Should the team ship version B?

**Pipeline mapping:**

| Stage       | Rain example (Sections 2–12)                 | A/B example (this section)                      |
|-------------|----------------------------------------------|-------------------------------------------------|
| Estimate    | MLE: $\hat{p} = \frac{8}{20} = 0.4$ (Sec 3)  | MLEs: $\hat{p}_A = 0.096$ , $\hat{p}_B = 0.112$ |
| Uncertainty | Posterior: Beta(11, 19) (Sec 4–5)            | Posteriors for $p_A$ , $p_B$                    |
| Test        | $H_0 : p \leq 0.3$ , p-value=0.228 (Sec 8–9) | $H_0 : p_A = p_B$ , test for lift               |
| Predict     | $P(\text{rain tomorrow}) = 0.367$ (Sec 11)   | $P(\text{user converts} \mid \text{version B})$ |
| Decide      | Move if $60q > 10$ (Sec 12)                  | Ship if $E[\text{gain}] > \text{rollout cost}$  |

**The structure is identical:** estimate parameters, quantify uncertainty, test hypotheses, predict outcomes, and decide based on expected loss.

### Tier 2: work through estimate, test, and decide

We model each version as an independent Bernoulli process:

$$X_A \sim \text{Binomial}(n_A, p_A), \quad X_B \sim \text{Binomial}(n_B, p_B).$$

#### Stage 1: Estimate (parallels Sections 2–3)

Point estimates (MLEs):

$$\hat{p}_A = \frac{96}{1000} = 0.096, \quad \hat{p}_B = \frac{112}{1000} = 0.112.$$

Lift estimate:

$$\hat{\Delta} = \hat{p}_B - \hat{p}_A = 0.016.$$

#### Stage 2: Test—Frequentist approach (parallels Sections 8–10)

Hypotheses:

$$\begin{aligned} H_0 : p_A &= p_B \quad (\text{no lift}), \\ H_1 : p_B &> p_A \quad (\text{positive lift}). \end{aligned}$$

Pooled proportion under  $H_0$ :

$$\hat{p}_{\text{pooled}} = \frac{96 + 112}{1000 + 1000} = 0.104.$$

Standard error under  $H_0$ :

$$\text{SE} = \sqrt{\hat{p}_{\text{pooled}}(1 - \hat{p}_{\text{pooled}}) \left( \frac{1}{n_A} + \frac{1}{n_B} \right)} = \sqrt{0.104 \times 0.896 \times 0.002} \approx 0.0136.$$

z-statistic:

$$z = \frac{\hat{p}_B - \hat{p}_A}{\text{SE}} = \frac{0.016}{0.0136} \approx 1.17.$$

p-value (one-sided):

$$\text{p-value} = P(Z \geq 1.17) \approx 0.121.$$

Conclusion: Do not reject  $H_0$  at  $\alpha = 0.05$ . The observed lift is not statistically significant.

95% confidence interval for lift:

$$[\hat{\Delta} - 1.96 \times \text{SE}_{\Delta}, \hat{\Delta} + 1.96 \times \text{SE}_{\Delta}] \approx [-0.011, 0.043],$$

where  $\text{SE}_{\Delta} = \sqrt{\hat{p}_A \frac{1-\hat{p}_A}{n_A} + \hat{p}_B \frac{1-\hat{p}_B}{n_B}} \approx 0.0137$ .

The CI includes 0, consistent with the non-significant test result. This confidence interval comes from a repeated-sampling procedure; it is not a statement that the fixed parameter has 95% probability of lying in this realized interval.

### Stage 2: Test—Bayesian approach (parallels Sections 4–5)

With uniform priors  $p_A \sim \text{Beta}(1, 1)$  and  $p_B \sim \text{Beta}(1, 1)$ , the posteriors are:

$$p_A \mid D \sim \text{Beta}(97, 905), \quad p_B \mid D \sim \text{Beta}(113, 889).$$

Posterior means:

$$E[p_A \mid D] = \frac{97}{1002} \approx 0.0968, \quad E[p_B \mid D] = \frac{113}{1002} \approx 0.1128.$$

To compute  $P(p_B > p_A \mid D)$ , we use Monte Carlo sampling (100,000 draws):

$$P(p_B > p_A \mid D) \approx 0.88.$$

This is below an illustrative organizational threshold of 0.95. Such a threshold is a policy choice, not a universal Bayesian rule.

95% credible interval for lift (via Monte Carlo):

$$[-0.011, 0.043].$$

This credible interval contains 95% of the posterior distribution of the lift under the stated priors and model.

### Stage 3: Decide (parallels Section 12)

Statistical evidence summary:

- Frequentist: p-value = 0.121 (not significant at 0.05)
- Bayesian:  $P(p_B > p_A \mid D) = 0.88$  (below 0.95 threshold)
- Point estimate: 1.6pp lift
- 95% CI includes 0

Business decision considerations:

1. **Effect size:** Is 1.6 percentage points meaningful? If each conversion is worth \$10 and monthly traffic is 100,000 visitors, the expected monthly gain is  $100000 \times 0.016 \times 10 = 16000$ , or \$16,000. Compare this to rollout cost.
2. **Downside risk:** The reported confidence interval has lower endpoint  $-1.1$  percentage points. This is not a literal worst case, but it shows that a negative lift is compatible with the data.
3. **Rollout cost:** If deploying version B requires 2 weeks of engineering effort (\$20,000), the expected gain must exceed this cost.

Illustrative evidence policies, none of which is an operational loss rule:

- **Corporate standard ( $p < 0.05$ ):** Do not ship ( $p = 0.121 > 0.05$ )
- **Pre-specified Bayesian threshold ( $P > 0.95$ ):** Investigate further ( $P = 0.88 < 0.95$ )
- **Positive lower confidence bound:** Do not ship (CI includes negative values)

**Context-dependent decision:** If rollout is low-cost (just a CSS change), easy to rollback, and **expected gain is positive**, the team might ship despite non-significance. If rollout requires major backend changes with downtime risk, the team would likely wait for more data.

### Tier 3: construct a decision with a time horizon

Let one conversion be worth \$10, monthly traffic be 100,000 visitors, and rollout cost be \$20,000. Using the estimated lift  $\hat{\Delta} = 0.016$ , the estimated monthly gross gain is

$$100000 \times 10 \times 0.016 = 16000 \text{ dollars.}$$

For a horizon of  $h$  months, the estimated net gain from shipping is

$$G(h) = 16000h - 20000 \text{ dollars.}$$

Thus  $G(1) = -4000$  dollars but  $G(2) = 12000$  dollars. The same statistical evidence can support different actions when the decision horizon changes. This is a substantive modeling choice, not a p-value threshold.

This calculation is risk-neutral and treats traffic, value per conversion, and lift as stable. A fuller decision integrates  $G(h)$  over posterior uncertainty and includes rollback or user-harm losses.

#### Capability Checkpoint: Transfer to A/B testing

Work the numbered questions Q41–Q43 in the end-of-sheet **Checkpoint Question Bank**. They are tiered as Easy, Medium, and Hard; try them before reading the solutions.

## 14. Closing synthesis: revise the opening rule

We began with a tempting rule: 8 rainy days out of 20 means an estimate of 0.4, and if the threshold is 30%, move indoors. That rule has the right instinct – data should influence action – but it **collapses four distinct questions** into one imprecise directive.

### The four questions, traced through the rainy-day example

**1. Estimate (Sections 2-7):** What is the rain probability parameter?

- MLE:  $\hat{p}_{\text{MLE}} = \frac{8}{20} = 0.400$
- Bayesian posterior:  $p \mid D \sim \text{Beta}(11, 19)$  with mean  $\frac{11}{30} \approx 0.367$
- Both methods give a point summary, but Bayesian estimation also quantifies uncertainty through the full posterior distribution

**2. Test (Sections 8-10):** Is the data consistent with a specific claim?

- Null hypothesis:  $H_0 : p \leq 0.3$  (event is safe enough for outdoor setup)
- Frequentist:  $p$ -value  $\approx 0.228 \rightarrow$  do not reject  $H_0$  at  $\alpha = 0.05$
- Bayesian:  $P(p > 0.3 \mid D) \approx 0.771 \rightarrow$  posterior evidence favors  $p > 0.3$ , but not overwhelmingly
- Testing answers “is the data surprising under  $H_0$ ?” It does NOT automatically tell you what action to take

**3. Predict (Section 11):** What will happen tomorrow?

- Plug-in prediction:  $P(\text{rain tomorrow}) = \hat{p}_{\text{MLE}} = 0.400$
- Posterior predictive:  $P(\text{rain tomorrow} \mid D) = E[p \mid D] = 0.367$
- The posterior predictive accounts for parameter uncertainty by averaging over the posterior; plug-in treats the MLE as certain
- The prediction feeds into the decision stage: we don’t experience  $p$  tomorrow, we experience rain or no rain

**4. Decide (Section 12):** Which action minimizes expected loss?

- Options: move indoors (cost 10) or stay outdoors (cost 0 if dry, 60 if rain)
- Using posterior predictive  $q = 0.367$ : expected loss for staying outdoors is  $60 \times 0.367 = 22.0$
- Decision: move indoors (expected loss  $10 < 22$ )
- Using plug-in  $q = 0.400$ : expected loss for staying outdoors is  $60 \times 0.400 = 24.0$
- Decision: still move indoors, but with a larger margin
- Key insight: the test said “do not reject  $H_0 : p \leq 0.3$ ” but the decision still says “move indoors” because the decision depends on loss ratios, not statistical significance

### What changes when we transfer to A/B testing (Section 13)

The same pipeline applies:

- 1. Estimate:** conversion rates  $\hat{p}_A = 0.096$ ,  $\hat{p}_B = 0.112$ , lift = 1.6 pp
- 2. Test:** frequentist  $p = 0.121$  (not significant); Bayesian  $P(p_B > p_A \mid D) = 0.88$
- 3. Predict:** if we ship version B, what conversion rate can we expect?
- 4. Decide:** does expected gain exceed rollout cost and downside risk?

The statistical test gives calibrated evidence, but the business decision also depends on:

- Effect size in revenue terms ( $1.6 \text{ pp} \times 100\text{k visitors} \times \$10/\text{conversion} = \$16\text{k/month}$ )
- Downside risk (95% CI lower bound is  $-1.1 \text{ pp}$ )
- Rollout cost and reversibility

All three illustrative evidence policies (corporate  $p < 0.05$ , pre-specified Bayesian  $P > 0.95$ , and a positive lower confidence bound) say “do not ship.” The explicit loss calculation can differ because it also includes value, cost, horizon, and reversibility.

## Why the distinctions matter

**Estimation is not testing.** You can estimate  $\hat{p} = 0.4$  and still fail to reject  $H_0 : p = 0.3$  because estimation gives a point summary, while testing asks whether the data are surprising under  $H_0$ .

**Testing is not deciding.** A non-significant test ( $p = 0.228$ ) does not mean “stay outdoors”; it means “the data are consistent with  $H_0$  at this error level.” The decision also depends on loss structure, costs, and constraints.

**Prediction is not the parameter.** Tomorrow’s rain is a random outcome generated according to  $p$ , not the parameter itself. Prediction requires integrating over parameter uncertainty (posterior predictive) or assuming the parameter is known (plug-in).

**A p-value is not a business metric.** The A/B test gives  $p = 0.121$ , but shipping decisions also require revenue impact, rollout costs, downside risk, and reversibility. Statistical significance is one input, not the full answer.

### Key Takeaway

Estimate when you need a parameter. Test when you need calibrated evidence against a claim. Predict when the next outcome matters. Decide when actions have losses. A mature statistical answer says **which of these questions it is answering** and **which assumptions make the answer meaningful**.

### Breadcrumb Prompt

#### Revise your starting rule

Rewrite the opening rule in four sentences, one per pipeline stage:

1. **Model and estimate:** “Under a Bernoulli model with prior  $\text{Beta}(3,7)$ , 8 rainy days out of 20 yields posterior  $\text{Beta}(11,19)$  with mean 0.367.”
2. **Test (if needed):** “Testing  $H_0 : p \leq 0.3$  gives  $p \approx 0.228$ ; do not reject. Under the  $\text{Beta}(11,19)$  posterior,  $P(p > 0.3 \mid D) \approx 0.771$ .”
3. **Predict:** “Posterior predictive rain probability tomorrow is 0.367.”
4. **Decide:** “Since expected outdoor loss  $60 \times 0.367 = 22.0 > 10$ , move indoors.”

## Checkpoint Question Bank

### Easy

1. In the campus event example, what is the difference between the parameter  $p$ , the estimator, and the estimate 0.4? **Solution.**

$p$  is the unknown rain probability in the model.  $\hat{p} = \sum_{i=1}^n \frac{x_i}{n}$  is the random rule. 0.4 is the realized value after observing 8 rainy days out of 20.

2. Name the four stages of the Week 3 inference-to-action pipeline. **Solution.**

Model, estimate, test or quantify uncertainty, and decide. Prediction may be inserted before decision when the next outcome matters.

3. What is the likelihood for  $m$  rainy days out of  $n$  under a Bernoulli model? **Solution.**

$L(d) = p^m (1-d)^{n-m}$ , ignoring the binomial coefficient if the exact sequence is treated as observed or if maximizing over  $p$ .

4. What is the Bernoulli MLE for  $p$ ? **Solution.**

$\hat{p}^{\text{MLE}} = \frac{m}{n}$ .

5. What posterior follows from prior  $p \sim \text{Beta}(a,b)$  and  $m$  rainy days out of  $n$ ? **Solution.**

$p \mid D \sim \text{Beta}(a+m, b+n-m)$ .

6. Which posterior summary minimizes squared-error loss? **Solution.**

The posterior mean.

7. What does a Type I error mean? **Solution.**

Rejecting the null hypothesis when the null is true.

8. What is a p-value not? **Solution.**

It is not the probability that the null hypothesis is true.

9. Under squared loss, which action should be chosen from a posterior distribution? **Solution.**

The posterior mean if the action is a point estimate; more generally, choose the action with smallest posterior expected loss.

10. Why is tomorrow's rain outcome not the same as the parameter  $p$ ? **Solution.**

$p$  is a model probability; tomorrow's outcome is a random event generated according to the model. Prediction concerns the outcome, not only the parameter.

### Medium

11. Derive the Bernoulli MLE by differentiating the log-likelihood. **Solution.**

For  $\ell(p) = m \log p + (n-m) \log(1-p)$ ,  $\frac{d}{dp} \ell(p) = \frac{m}{p} - \frac{n-m}{1-p}$ . Setting to zero gives  $m(1-p) = (n-m)p$ , hence  $\hat{p} = \frac{m}{n}$ .

12. For prior  $\text{Beta}(3,7)$  and data  $m=8, n=20$ , compute the posterior mean. **Solution.**

The posterior is  $\text{Beta}(11, 19)$ , so the posterior mean is  $\frac{11}{30} \approx 0.367$ .

13. Explain why MAP and the posterior mean can differ. **Solution.**

MAP is the posterior mode; the posterior mean averages over the whole distribution. They differ when the posterior is skewed or constrained.

14. Show why Gaussian MLE for known variance reduces to least squares. **Solution.**

The Gaussian log-likelihood equals a constant minus  $\frac{1}{2} \sum_i (X_i - \mu)^2$ . Maximizing it is equivalent to minimizing the sum of squared deviations.

15. State the MSE decomposition. **Solution.**

$$\text{MSE}(\theta) = \text{Var}(\theta) + \text{Bias}(\theta)^2.$$

16. Why is the MLE variance estimator with denominator  $n$  biased downward? **Solution.**

The sample mean is fitted from the same data and reduces residuals. One degree of freedom is used to estimate  $\mu$ , yielding expectation  $\left(\frac{n-1}{n}\right) \sigma^2$ .

17. For a one-sided rainy-day test with null hypothesis  $p=0.3$ , what is the p-value after observing  $M=8$  out of 20? **Solution.**

$$\sum_{k=8}^{20} \binom{20}{k} 0.3^k 0.7^{20-k} \approx 0.228, \text{ so the test does not reject at } \alpha = 0.05.$$

18. Define the generalized likelihood-ratio statistic. **Solution.**

$$\Lambda(D) = \sup_{\theta \in \Theta_0} \frac{L(\theta)}{L(\hat{\theta})}. \text{ Small values indicate evidence against the null.}$$

19. If moving indoors costs 10 and rain outdoors costs 60, at what predictive rain probability should the team move indoors? **Solution.**

$$\text{Move indoors if } 60p > 10, \text{ i.e. } p > \frac{1}{6}.$$

20. In the A/B transfer example, why is “B has more conversions” not enough to ship B? **Solution.**

It ignores sampling noise, effect size, uncertainty, rollout cost, and downside risk.

## Hard

21. Give an example of an unbiased estimator that is worse than a biased estimator for a practical task. **Solution.**

For Gaussian mean estimation,  $X_1$  is unbiased for  $\mu$  but discards the rest of the sample and has larger variance than  $\bar{X}$ . A shrinkage estimator can be biased but lower MSE in some settings.

22. Explain the difference between a p-value and posterior probability of the null. **Solution.**

The p-value is  $P(\text{data as extreme} \mid H_0)$ , a sampling probability under the null. Posterior probability is  $P(H_0 \mid \text{data})$ , requiring a prior and Bayes' rule. They condition in opposite directions.

23. Suppose no rain is observed in five days. Compare MLE and Bayesian posterior mean under uniform prior. **Solution.**

MLE is  $\frac{5}{6} = 0$ . The posterior is  $\text{Beta}(1, 6)$  and the posterior mean is  $\frac{1}{7}$ , avoiding the zero-probability plug-in prediction.

24. Why can a statistically significant result fail to justify an action? **Solution.**

Statistical significance addresses evidence against a null at a chosen error rate. Action requires effect size, utility/loss, constraints, and risk tolerance.

25. Why can failing to reject a null still justify a cautious action? **Solution.**

The test may have low power or the loss of being wrong may be severe. Decision thresholds depend on expected loss, not only rejection at level  $\alpha$ .

26. In what sense is MLE “asymptotically good,” and what caveat should accompany that statement? **Solution.**

Under regularity conditions and correct model specification, MLE is often consistent, asymptotically normal, and asymptotically efficient. These are large-sample, model-dependent claims, not finite-sample guarantees.

27. Describe how the posterior predictive distribution differs from plug-in prediction. **Solution.**

Plug-in prediction uses one estimate  $\hat{p}$ . Posterior prediction integrates  $P(X^{n+1}|D)$  over  $P(D)$ , accounting for parameter uncertainty inside the model.

28. Construct a null and alternative hypothesis for the claim “version B improves conversion.” **Solution.**

One common formulation is  $H_0: p_B \leq p_A$  versus  $H_1: p_B > p_A$ , or the simple no-lift null  $H_0: p_B = p_A$  versus one-sided alternative  $p_B > p_A$ .

29. Explain why cross-entropy is a loss for probability predictions rather than point estimates of a real-valued parameter. **Solution.**

Cross-entropy scores an entire predictive distribution or probability vector. It penalizes assigning low probability to the observed class and is minimized by calibrated probabilities under the true distribution.

30. Rewrite the opening rule into a mature statistical rule. **Solution.**

A strong answer names the Bernoulli event-day model, estimates or updates  $p$ , checks whether data contradict a policy threshold if needed, computes predictive rain probability, and chooses indoors/outdoors by expected loss rather than a fixed 50% or raw sample-frequency rule.

31. For a Bernoulli likelihood-ratio test, what does a small Lambda value mean? **Solution.**

It means the best fit under the null has much smaller likelihood than the unrestricted best fit, so the data provide evidence against the null model.

32. Why does the likelihood-ratio formula need a boundary caveat when  $m=0$  or  $m=n$ ? **Solution.**

The unrestricted MLE is then at  $p = 0$  or  $p = 1$ , so terms such as  $\binom{n}{m} p^m (1-p)^{n-m}$  require the usual limiting convention, including 0<sup>0</sup>-style factors from empty products.

33. Why should small-sample Bernoulli testing prefer exact binomial reasoning over a large-sample chi-square shortcut? **Solution.**

Large-sample approximations rely on regularity and enough data. With small samples or boundary behavior, exact binomial probabilities better match the stated model.

34. Define posterior expected loss for an action. **Solution.**

$p(a|D) = E[T(X, a) | D]$ , where  $X$  is the future outcome or target of action and the expectation uses posterior or predictive uncertainty.

35. If moving indoors costs 15 and rain outdoors costs 75, what predictive rain probability makes moving indoors preferable? **Solution.**

Move indoors if  $75q > 15$ , so  $q > \frac{15}{75} = 0.2$ .

36. Name the posterior summaries associated with squared loss and absolute loss. **Solution.**

Squared-error loss is minimized by the posterior mean. Absolute-error loss is minimized by a posterior median.

37. Why can rejecting the null hypothesis still fail to justify moving indoors? **Solution.**

Rejection is evidence against a statistical claim at a chosen error level. The action also depends on moving cost, rain damage, uncertainty, constraints, and risk tolerance.

38. In the Beta-Bernoulli model, what is the one-step posterior predictive rain probability? **Solution.**

$P(X^{n+1} = 1 | D) = \frac{a+b+n}{a+m}$ , assuming the future Bernoulli event is conditionally independent of the past given  $p$ .

39. Why is plug-in prediction not the same as posterior predictive prediction? **Solution.**

Plug-in prediction uses one fitted value such as  $\hat{p}$ . Posterior predictive prediction averages  $P(X^{n+1} | p)$  over the posterior distribution of  $p$ .

40. Name one model failure that posterior predictive averaging does not fix. **Solution.**

Examples include seasonality, dependence among weather days, event-selection bias, changing forecast systems, or a wrong Bernoulli/i.i.d. assumption.

41. For A/B data A: 96/1000 and B: 112/1000, what is the estimated conversion lift? **Solution.**

$\hat{p}_A = 0.096$ ,  $\hat{p}_B = 0.112$ , so the estimated lift  $\hat{p}_B - \hat{p}_A = 0.016$ , or 1.6 percentage points.

42. In the A/B transfer pipeline, separate estimate, test, and decision in one sentence each. **Solution.**

Estimate: compute the conversion-rate lift. Test: ask whether the observed lift is distinguishable from sampling noise under a no-lift null. Decision: ship only if expected gain exceeds rollout costs and downside risks.

43. Give one rollout cost or risk that an A/B test p-value does not include. **Solution.**

Examples include engineering cost, user harm, long-term retention risk, fairness concerns, monitoring cost, opportunity cost, or risk that the experiment population differs from the rollout population.